

Saif Al-Din Samy

HPC Systems Intern

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October Gardens, Giza

Education

Cairo University | GPA: 3.18

Sep 2022 - Jul 2026

Bachelor's of Computer Science | Computer Science

Giza, Egypt

Experience

HPC R&D Assistant | Cairo University

Apr 2025 - Present

- Member of the High Performance Computing research group under faculty supervision at Cairo University.
- Built and configured two cluster architectures: a manually-configured Beowulf cluster and an xCAT-provisioned cluster with FreeIPA for identity management and SLURM for job scheduling.
- Conducted initial experiments on parallel workload distribution across compute nodes using MPI.

Fullstack Developer Intern | Cloudjet Innovations

Jul 2025 - Sep 2025

- Co-developed LearnMate, a study platform serving students with tools for note-taking, PDF reading, and AI-assisted learning.
- Deployed and hosted the application on a Linux server, gaining hands-on experience with server configuration and management.
- Led a 3-person team, coordinating development tasks and presenting the final product on pitch day.

Freelance Fullstack Developer

Feb 2024 - Apr 2025

- Developed and delivered full-stack websites for clients, including a data management dashboard with real-time statistics for improved decision-making.

Skills

Languages: C, Go, Rust, Bash

HPC & Parallel Computing: MPI, OpenMP, xCAT, SLURM, FreeIPA, Cluster Deployment (Beowulf, Master/Compute)

Tools: Git, Linux System Administration, Linux Networking (nftables, tcpdump), Docker/Podman, LXC, QEMU/KVM, Nix/NixOS/Nix Containers

Projects

xCAT-Provisioned HPC Cluster | Linux, Bash, ssh, xCAT, FreeIPA, NFS, SLURM

2026

- Built a homogeneous cluster consisting of a login/management node and 2 compute nodes.
- Configured node networking, user synchronization, and shared storage via NFS, and job scheduling across compute nodes.

HiPer-FC HPC Cluster Rebuild | Linux, Bash, MPI, OpenMP

2025

- Rebuilt HiPer-FC, a Beowulf cluster with 5 nodes during my time at Cairo University's HPC R&D lab.
- Tested the cluster by developing and running parallel MPI/OpenMP applications across the 5 nodes.

Home Lab / Self-Hosted Infrastructure | NixOS, systemd, nftables, Podman, Nix flakes

2025 — Present

<https://github.com/owlenz/dotfiles>

- Designed and maintain a multi-host NixOS infrastructure managed declaratively via Nix flakes, including a home server and personal workstations.
- Deployed and operate a containerized media/services stack using Podman with rootless OCI containers, systemd service units, and declarative secrets management via sops-nix.
- Manage system-level Linux administration: user/group permissions, ACLs, systemd unit design, and binary-cache/build optimization for reproducible deployments.

CTTP | C, Unix Sockets

Mar 2025

<https://github.com/owlenz/cttp>

- Developed a lightweight HTTP server in C, utilizing Unix sockets for efficient connections and POSIX threads for enhanced load handling and concurrency.
- Handles standard HTTP methods (GET, POST, PUT, etc.).